(1) a) $M\ddot{q} + C\dot{q} + Kq = f(t)$

1) mass-spring-damper system in mechanical engineering

~~electrical circuit~~ $\Rightarrow$ $M \to$ mass, $C \to$ damping, $K \to$ spring stiffness

$q \to$ displacement

example: vibration damping systems

car suspension

2) electrical circuit in electrical engineering

$\Rightarrow$ $M \to L$ (inductance), $C \to R$ (resistance), $K \to C^{-1}$ (capacitance)

$q \to$ charge / current

3) flight dynamics system (rotational) in aerospace engineering

$\Rightarrow$ $M \to I$ (moment of inertia), $C \to$ aerodynamic damping,

$K \to$ aerodynamic stiffness (stability)

$q \to \theta$ ~~[illegible]~~ (angular displacement)

b) <u>for homogeneous version (i.e. $M\ddot{q} + C\dot{q} + Kq = 0$):</u>

$\to$ characteristic equation

$$Mr^2 + Cr + K = 0$$

<u>for non-homogeneous (i.e. $f(x) \neq 0$):</u>

$\to$ method of undetermined coefficients

$\to$ method of variation of parameters

$\to$ Laplace transforms

$\to$ numerical methods

c) <u>Explicit methods:</u>

→ future state of system computed from past + current state values

> conditionally stable → smaller Δt required

> computationally cheaper in general (direct computations)

> require special 'startup' for first step (not self-starting)

- for MDOF system (M, C, K are matrices), Δt depends on the max natural frequency of the system. (i.e. $\Delta t \leq 2/\omega_{max}$)

<u>Implicit methods:</u>

→ does not require past time state values to solve future values

> more stable in general at larger time steps → Δt more lenient

> more expensive per time step (solves system of equations)

> only requires initial conditions, but is self-starting

- For multi-DOF system (i.e. Houbolt), forms eff. stiffness matrix K, then solves a linear system at each time step for future displacement. This allows for higher stability at larger Δt.

(2) (Good student)

→ see "P_2" in files

One eigenvalue of K is negative → system instability

↳ (see code output in console)

③ (grad. only)

eqn (1) $\rightarrow$ $\overset{(1)}{\frac{\partial^2 u}{\partial x_1^2}} + \overset{(2)}{\frac{\partial^2 u}{\partial x_2^2}} = \alpha \overset{(3)}{\frac{\partial^2 u}{\partial t^2}}$ 2D space (x_1, x_2) time (t)

(wave eqn)

so assume seperable soln. of form $u(x_1, x_2, t) = \bar{X}(x_1)\,\bar{Y}(x_2)\,T(t)$

so term ① $\Rightarrow \bar{X}''\bar{Y}T$

② $\Rightarrow \bar{X}\bar{Y}''T$ $\Rightarrow$ Pull out common $\bar{X}\bar{Y}T$, plug into (1)

③ $\Rightarrow \alpha\bar{X}\bar{Y}T''$

(1) $\Rightarrow \cancel{\bar{X}\bar{Y}T}\left(\frac{\bar{X}''}{\bar{X}} + \frac{\bar{Y}''}{\bar{Y}}\right) = \cancel{\bar{X}\bar{Y}T}\left(\alpha\frac{T''}{T}\right)$ (2)

(2) $\Rightarrow \underbrace{\frac{\bar{X}''}{\bar{X}} + \frac{\bar{Y}''}{\bar{Y}}}_{\text{only dependent on spatial}} = \underbrace{\alpha\frac{T''}{T}}_{\text{only dependent on temporal}} = \underbrace{0}_{\text{constant}}$ $\Rightarrow$ for the sum of 2 functions that each only depend on seperate variables, to = a constant, both functions must be constant, since constant is zero, the functions are also equal

$\therefore \Rightarrow \frac{\bar{X}''}{\bar{X}} + \frac{\bar{Y}''}{\bar{Y}} = \lambda$ (3) and $\alpha\frac{T''}{T} = \lambda$ (4), where λ is a constant

so time eqn (4) $\Rightarrow \left[T'' + \frac{\lambda}{\alpha}T = 0\right]$

(note: λ absorbs negative sign)

seperate space further $\Rightarrow$ (3) let $\rightarrow \frac{\bar{X}''}{\bar{X}} = \gamma$ + $\frac{\bar{Y}''}{\bar{Y}} = \lambda - \gamma$, where γ is another constant

so $\Rightarrow$ spatial eqn (3) and $\left[\begin{array}{l}\bar{X}'' + \gamma\bar{X} = 0 \\ \bar{Y}'' + (\lambda-\gamma)\bar{Y} = 0\end{array}\right]$ by substitution of above eqns into (3)

eqn (2) $\rightarrow \frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} = \alpha\frac{\partial u}{\partial t}$ Same spatial equations as eqn (1), but time only has 1st order diff.

(heat eqn)

$\therefore \Rightarrow$ time eqn $\Rightarrow \left[T' + \frac{\lambda}{\alpha}T = 0\right]$

b)

(both) IC: 1) $u(x_1, x_2) = 0$ at $t=0$

$\rightarrow u(x_1, x_2, 0) = 0$

(both) BC: 1) $u(0, x_2, t) = 10$ mm

2) $u(L, x_2, t) = 1$ mm

3) $u(x_1, 0, t) = 0$ mm

4) $u(x_1, W, t) = 0$ mm

(4) (bonus)

$$f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + x_2^2 + 2x_1x_2$$

$$\bar{X} = [0, 0]$$

see P_4 ode for results

optimal point $\Rightarrow (x_1^*, x_2^*) = (-0.499..., 1.499...)$

minimum value $\Rightarrow f^* = -1.250...$

analytical solu $\Rightarrow (X_1^*, X_2^*) = (-1.000, 1.500)$

$f^* = -1.25$

errors $\Rightarrow$ $e_{x^*} \approx 1.01e^{-6}$

$e_{f^*} \approx 2.22e^{-16}$